

MATH1013 Assignment 2

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Theorems

Herein, *Calculus* and *Spivak* will refer to the book *Calculus, Third Edition* by *Michael Spivak* published in 1994, printed in 2009. **ISBN 978-0-521-86744-3.**

All theorems published in *Spivak* and relied upon for this assignment are printed below in order of appearance.

1. (Theorem 10-4, pg. 167) If f and g are differentiable at a , then

$$(f \cdot g)'(a) = f'(a)g(a) + g'(a)f(a) \quad (1)$$

2. (Theorem 10-9, pg. 176) If g is differentiable at a , and f is differentiable at $g(a)$, then $f \circ g$ is differentiable at a , and

$$(f \circ g)'(a) = f'(g(a)) \cdot g'(a) \quad (2)$$

3. (Theorem 10-5, pg. 167) If $g(x) = c \cdot f(x)$, and f is differentiable at a , then g is differentiable at a , and

$$g'(a) = c \cdot f'(a) \quad (3)$$

4. (Theorem 10-7, pg. 169) If g is differentiable at a , and $g(a) \neq 0$, then $\frac{1}{g}$ is differentiable at a , and

$$\left(\frac{1}{g}\right)'(a) = \frac{-g'(a)}{(g(a))^2} \quad (4)$$

5. (Theorem 5-2, pg. 102) If $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$, then

$$\lim_{x \rightarrow a} (f + g)(x) = l + m \quad (5)$$

$$\lim_{x \rightarrow a} (f \cdot g)(x) = l \cdot m \quad (6)$$

6. (Theorem 6-1, pg. 114) If f and g are continuous at a , then

$$f + g \text{ is continuous at } a \quad (7)$$

7. (Theorem 7-1, pg. 120) If f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$, then there is some $x \in [a, b]$ such that

$$f(x) = 0 \quad (8)$$

8. (Theorem 11-9, pg. 201) Suppose that $\lim_{x \rightarrow a} f(x) = 0$, and $\lim_{x \rightarrow a} g(x) = 0$, and suppose that the $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists. Then the $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ exists and

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \quad (9)$$

9. (Theorem 10-6, pg. 168) If $f(x) = x^n$ for some natural number n , then

$$f'(a) = na^{n-1} \quad \forall a \quad (10)$$

1 Exercise 1

It is given that $\forall x \in \mathbb{R}$ the first derivative $(\cos(x))' = -\sin(x)$.

For $\alpha, \beta \in (0, +\infty)$ let $f : [0, 1) \rightarrow \mathbb{R}$ be defined:

$$f(x) := \begin{cases} x^\alpha \cos\left(\frac{1}{x^\beta}\right) & x \in (0, 1) \\ 0 & x = 0 \end{cases} \quad (11)$$

1.1

Compute $f'(x)$ for every $x \in (0, 1)$

1.1.1 Solution

For $x \in (0, 1)$, $f(x) = x^\alpha \cos\left(\frac{1}{x^\beta}\right)$ from eq. (11).

Define $g(x) := x^\alpha$. Define $h(x) := \left(\frac{1}{x^\beta}\right)$.

Then,

$$f(x) = g(x) \cos(h(x)) \quad (12)$$

Thus, from eq. (1) and eq. (2),

$$f'(x) = g'(x) \cos(h(x)) + g(x) (\cos'(h(x)) \cdot h'(x)) \quad (13)$$

First, $g'(x)$. Assuming $g(x) > 0$ which is given:

$$g(x) = x^\alpha$$

Take the natural log of both sides: $\ln(g(x)) = \ln(x^\alpha)$

$$\ln(g(x)) = \alpha \ln(x)$$

Differentiate both sides w.r.t. x : $(\ln(g(x)))' = (\alpha \ln(x))'$

From eq. (2) and eq. (3): $\ln'(g(x)) \cdot g'(x) = \alpha \ln'(x)$

$$\text{So: } g'(x) = \frac{\alpha \ln'(x)}{\ln'(g(x))} \quad (14)$$

The derivative of the natural logarithm:

$$\begin{aligned} \ln'(x) &= \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln(x)}{h} \text{ from the definition of the derivative.} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \ln\left(1 + \frac{h}{x}\right) \end{aligned}$$

Let $u = \frac{h}{x} \implies h = xu$

$$\begin{aligned} \ln'(x) &= \lim_{u \rightarrow 0} \frac{1}{xu} \ln(1+u) \\ &= \frac{1}{x} \lim_{u \rightarrow 0} \frac{1}{u} \ln(1+u) \\ &= \frac{1}{x} \lim_{u \rightarrow 0} \ln(1+u)^{\frac{1}{u}} \\ &= \frac{1}{x} \ln\left(\lim_{u \rightarrow 0} (1+u)^{\frac{1}{u}}\right) \end{aligned}$$

We do not need to evaluate this limit since it is, by definition, the base of the natural logarithm, e , and $\ln(e) = 1$ from the definition of the natural logarithm. So:

$$\ln'(x) = \frac{1}{x} \quad (15)$$

From eq. (14) and eq. (15):

$$g'(x) = \frac{\frac{\alpha}{x}}{\frac{1}{g(x)}} = \frac{\alpha g(x)}{x} = \alpha x^{\alpha-1} \quad (16)$$

Now, $h'(x)$:

Define $y(x) := x^\beta$ such that $h(x) = \frac{1}{y(x)}$, then $h'(x) = \left(\frac{1}{y}\right)'(x)$ from eq. (4). Then:

$$h'(x) = \frac{-y'(x)}{(y(x))^2} \quad (17)$$

From eq. (16):

$$h'(x) = \frac{-\beta x^{\beta-1}}{x^{2\beta}} = -\beta x^{-\beta-1} \quad (18)$$

Since, the derivative of cosine is given as negative sine, eq. (13) becomes, from eq. (16) and eq. (18)

$$\alpha x^{\alpha-1} \cos\left(\frac{1}{x^\beta}\right) + \beta x^{\alpha-\beta-1} \sin\left(\frac{1}{x^\beta}\right) \quad (19)$$

1.2

Determine all the values $\alpha, \beta \in (0, +\infty)$ such that

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x} \text{ exists.}$$

1.2.1 Solution

From eq. (11):

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0^+} \frac{x^\alpha \cos(x^{-\beta}) - 0}{x} = \lim_{x \rightarrow 0^+} x^{\alpha-1} \cos\left(\frac{1}{x^\beta}\right) \quad (20)$$

Case 1: $\alpha = 1 \implies \alpha - 1 = 0 \implies x^{\alpha-1} = 1$. The limit becomes $\lim_{x \rightarrow 0^+} \cos\left(\frac{1}{x^\beta}\right)$ which does not exist.

Proof:

Assume for contradiction that the limit does exist and equals l , then, $\forall \epsilon > 0, \exists \delta > 0$, such that if $0 < |x - a| < \delta, |f(x) - l| < \epsilon$. Choose a sufficiently large natural number N , such that $(\frac{1}{2\pi n})^{1/\beta}$ is less than δ . Then there will be a point $a = (\frac{1}{2\pi n})^{1/\beta}$ where $\cos(a) = 1$ and a point $b = (\frac{1}{2\pi n+1})^{1/\beta} < a < \delta$ where $\cos(b) = -1$.

Choose $\epsilon = 1$. From the definition of the limit, $|f(a) - l| < 1 \implies |1 - l| < 1$, and $|f(b) - l| < 1 \implies |-1 - l| < 1$.

From the triangle inequality:

$|1 - (-1)| = |(1 - l) + (l + 1)| \leq |1 - l| + |1 + l| < 1 + 1 < 2 \implies 2 < 2$. Which is a contradiction. Thus, the limit does not exist. $\square$

Case 2: $0 < \alpha < 1 \implies \alpha - 1 < 0$ The term $x^{\alpha-1}$ goes to infinity as x goes to 0, and since the $\cos(\frac{1}{x^\beta})$ term oscillates between 0 and 1 and does not approach any limit at 0 as shown above, the limit diverges and does not exist.

Proof:

A function f approaches $+\infty$ at a if $\forall M \in \mathbb{R}, \exists \delta > 0$ such that $\forall x$, if $0 < |x - a| < \delta$, $f(x) > M$. Let $p = \alpha - 1$. Since $0 < \alpha < 1$, $-1 < p < 0$. Let $q = -p$, then $0 < q < 1$. $x^p = x^{-q} = \frac{1}{x^q}$. If we choose $\delta = (\frac{1}{M})^{1/q}$, then for all $0 < x < \delta$, $x^p > M$.

To verify, choose M to be arbitrary, and $\delta = (\frac{1}{M})^{1/q} = (\frac{1}{M})^{1/(1-\alpha)}$. Then $x < (\frac{1}{M})^{1/(1-\alpha)}$, so $x^{1-\alpha} < \frac{1}{M} \implies x^{-(1-\alpha)} > M \implies x^{\alpha-1} > M$. $\square$

Case 3: $\alpha > 1 \implies x^{\alpha-1} > 0$ when $x > 0$. $x^{\alpha-1}$ goes to 0 as x goes to 0 from the right. Since $0 \leq |\cos(\frac{1}{x^\beta})| \leq 1$, $|x^{\alpha-1} \cos(\frac{1}{x^\beta})| \leq x^{\alpha-1}$ which goes to 0 as $x \rightarrow 0^+$, so the whole function goes to 0 as $x \rightarrow 0^+$ since the limit of products is the product of limits (eq. (6)).

Proof:

From the definition of a limit, $\forall \epsilon > 0, \exists \delta > 0$, such that, if $0 < x < \delta$, $|f(x) - 0| < \epsilon$. $|f(x)|$ is bounded by $x^{\alpha-1}$ since $0 \leq |\cos(x)| \leq 1, \forall x$. So if we choose $\epsilon > x^{\alpha-1}$, $x < \epsilon^{\frac{1}{\alpha-1}}$. Thus, $\delta = \epsilon^{\frac{1}{\alpha-1}}$.

To verify, let $\epsilon > 0$ be given, and choose $\delta = \epsilon^{\frac{1}{\alpha-1}}$. For all x such that $0 < x < \delta$, $x < \epsilon^{\frac{1}{\alpha-1}}$. Raise both sides to the power of $\alpha - 1$, $x^{\alpha-1} < \epsilon$. Since $|x^{\alpha-1} \cos(\frac{1}{x^\beta})| \leq x^{\alpha-1} < \epsilon$, it follows that $|f(x)| < \epsilon$. $\square$

Considering all the three cases together, the only values of α and $\beta \in (0, +\infty)$ where the limit exists are $\alpha \in (1, +\infty), \beta \in (0, +\infty)$.

1.3

Determine all values of $\alpha, \beta \in (0, +\infty)$ such that

$$\lim_{x \rightarrow 0^+} f'(x) \text{ exists.}$$

1.3.1 Solution

The derivative of $f(x), f'(x)$ (eq. (19)) is

$$\underbrace{\alpha x^{\alpha-1} \cos\left(\frac{1}{x^\beta}\right)}_{\text{Term One}} + \underbrace{\beta x^{\alpha-\beta-1} \sin\left(\frac{1}{x^\beta}\right)}_{\text{Term Two}}$$

Each term behaves similarly and will oscillate between -1 and 1 as x goes to 0 from the right. I have already proven above that for the limit of term one to exist, $\alpha > 1$. I have also already proven that if $\alpha - \beta - 1 < 0$, the limit will not exist since $x^{\alpha-\beta-1}$ will diverge. So $\alpha - \beta - 1$ must at least be zero. Since the sum of limits is the limit of sums (eq. (5)), we can consider each term separately.

Case 1: $\alpha - \beta - 1 = 0$ and the limit of term two becomes $\beta \sin\left(\frac{1}{x^\beta}\right)$. The $\lim_{x \rightarrow 0^+} \beta \sin\left(\frac{1}{x^\beta}\right)$ does not exist for the same reason the limit of cosine of the same term does not exist.

Assume for contradiction that the limit does exist and equals l , then, $\forall \epsilon > 0, \exists \delta > 0$, such that if $0 < |x - a| < \delta$, $|f(x) - l| < \epsilon$. Choose a sufficiently large natural number N , such that $(\frac{1}{2\pi n})^{1/\beta}$ is less than δ . Then there will be a point $a = (\frac{1}{2\pi n + \pi/2})^{1/\beta}$ where $\beta \sin(a) = \beta$ and a point $b = (\frac{1}{2\pi n + 3\pi/2})^{1/\beta} < a < \delta$ where $\beta \sin(b) = -\beta$.

Choose $\epsilon = \beta$. From the definition of the limit, $|f(a) - l| < \beta \implies |\beta - l| < \beta$, and $|f(b) - l| < \beta \implies |-\beta - l| < \beta$.

From the triangle inequality:

$|\beta - (-\beta)| = |(\beta - l) + (l + \beta)| \leq |\beta - l| + |\beta + l| < \beta + \beta < 2\beta \implies 2 < 2$ (since $\beta > 0$). Which is a contradiction. Thus, the limit does not exist. $\square$

Case 2: $\alpha - \beta - 1 > 0$. $x^{\alpha-\beta-1}$ goes to 0 as x goes to 0 from the right. Since $0 \leq |\sin(\frac{1}{x^\beta})| \leq 1$, $|\beta x^{\alpha-\beta-1} \sin(\frac{1}{x^\beta})| \leq \beta x^{\alpha-\beta-1}$ which goes to 0 as $x \rightarrow 0^+$, so the whole function goes to 0 as $x \rightarrow 0^+$ since the limit of products is the product of limits (eq. (6)).

Proof:

From the definition of a limit, $\forall \epsilon > 0, \exists \delta > 0$, such that, if $0 < x < \delta$, $|f(x) - 0| < \epsilon$. $|f(x)|$ is bounded by $\beta x^{\alpha-\beta-1}$ since $0 \leq \sin(x) \leq 1, \forall x$. So if we choose $\epsilon > \beta x^{\alpha-\beta-1}$, $x < (\frac{\epsilon}{\beta})^{\frac{1}{\alpha-\beta-1}}$. Thus, $\delta = (\frac{\epsilon}{\beta})^{\frac{1}{\alpha-\beta-1}}$.

To verify, let $\epsilon > 0$ be given, and choose $\delta = (\frac{\epsilon}{\beta})^{\frac{1}{\alpha-\beta-1}}$. For all x such that, if $0 < x < \delta$, $x < (\frac{\epsilon}{\beta})^{\frac{1}{\alpha-\beta-1}}$. Raise both sides to the power of $\alpha - \beta - 1$, so $x^{\alpha-\beta-1} < \frac{\epsilon}{\beta} \implies \beta x^{\alpha-\beta-1} < \epsilon$. Since $|\beta x^{\alpha-\beta-1} \sin(\frac{1}{x^\beta})| \leq \beta x^{\alpha-\beta-1} < \epsilon$, this implies $|f(x)| < \epsilon$.

Considering the solution to 1.2, as well as the cases where the limit of the second term exists, we require that $\alpha - \beta - 1 > 0$ and $\alpha > 1$. So, $\alpha > \beta + 1$ which requires $\beta > 0$ to satisfy $\alpha > 1$ for the limit of the derivative to exist as x goes to 0 from the right.

2 Exercise 2

We say that $f : \mathbb{R} \rightarrow \mathbb{R}$ is even if for every $x \in \mathbb{R}$ it holds that $f(x) = f(-x)$. Also, we say that f is odd if for every $x \in \mathbb{R}$ it holds that $f(x) = -f(-x)$. Prove that if f is differentiable everywhere in $\mathbb{R}$ then

$$f' \text{ is even if } f \text{ is odd, and that } f' \text{ is odd if } f \text{ is even.}$$

Assume that f is k -times differentiable. What can be said about $f^{(k)}$?

2.1 Solution

Suppose that f is even, so $f(x) = f(-x)$. Differentiating both sides yields $f'(x) = f'(-x) \cdot (-x)'$ from eq. (2). The derivative of $-x$

$$\begin{aligned} (-x)' &= \lim_{h \rightarrow 0} \frac{-(x+h) - (-x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-x - h + x}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{h} \\ &= -1 \end{aligned}$$

So $f'(x) = -f'(-x)$ which is odd by definition. So the first derivatives of even functions are odd. $\square$

Suppose that g is odd, so $g(x) = -g(-x)$. Differentiating both sides yields $g'(x) = (-1)g'(-x)(-x)'$ from eq. (2) and eq. (3). The derivative of $-x$ has already been shown to be -1 , so $g'(x) = (-1)(-1)g'(-x) = g'(-x)$ which is even by definition. So the first derivative of odd functions are even. $\square$

It is a direct corollary of the above two statements that, if the first derivative of an even function is odd, and the first derivative of an odd function, even, then the second derivative of an even function is even, and the second derivative of an odd function, odd. Thus, for the k 'th derivative of f , if k is even, then $f^{(k)}$ will be the same as f in terms of being even or odd, and if k is odd, then $f^{(k)}$ will be opposite f in terms of being even or odd.

3 Exercise 3

Suppose that f is a continuous function on the reals and

$$\lim_{x \rightarrow +\infty} \frac{f(x)}{x^{27}} = \lim_{x \rightarrow -\infty} \frac{f(x)}{x^{27}} = 0$$

Prove that there exists x_0 such that $x_0^{27} + f(x_0) = 0$

3.1 Solution

Define $g(x) := x^{27} + f(x)$.

Since the $\lim_{x \rightarrow +\infty} \frac{f(x)}{x^{27}} = 0$, there exists some $M \in \mathbb{R}^+$ such that for all $x > M$, $\left| \frac{f(x)}{x^{27}} \right| < \frac{1}{2}$ from the definition of a limit (choosing $\epsilon = 1/2$). It follows that $|f(x)| < \frac{x^{27}}{2}$. Take some $a > M > 0$, then $g(a) = a^{27} + f(a) > a^{27} - \frac{a^{27}}{2} = \frac{a^{27}}{2} > 0$.

Since the $\lim_{x \rightarrow -\infty} \frac{f(x)}{x^{27}} = 0$, there exists some $M' \in \mathbb{R}^+$ such that for all $x < -M'$, $\left| \frac{f(x)}{x^{27}} \right| < \frac{1}{2} \implies |f(x)| < \frac{|x|^{27}}{2}$. Take some $b < -M' < 0$, then $g(b) = b^{27} + f(b) < b^{27} + \frac{|b|^{27}}{2} = b^{27} - \frac{b^{27}}{2} < 0$. Note that $|b|^{27} = -b^{27}$ since 27 is odd.

Since x^{27} is continuous and f is continuous, g is also continuous on the reals (eq. (7)) and so is continuous on $[b, a]$. Since $f(b) < 0 < f(a)$ then by eq. (8) there is some $x_0 \in [b, a]$ where $g(x_0) = 0$ and therefore some point where $x_0^{27} + f(x_0) = 0$. $\square$

4 Exercise 4

Let f and g be bounded functions on the reals. Suppose that f and g are uniformly continuous and for all $x \in \mathbb{R}$ let

$$h(x) := f(x)g(x) \quad (21)$$

Prove that h is uniformly continuous. If we drop the assumption that f and g are bounded, is h necessarily uniformly continuous?

4.1 Solution

It is given that f and g are bounded, so there must be some $N, M \in \mathbb{R}$ such that $|f(x)| \leq N$, and $|g(x)| \leq M$. To show h is uniformly continuous, we must show that $\forall \epsilon > 0, \exists \delta > 0$ such that $\forall x, y \in \mathbb{R}$, if $|x - y| < \delta$, $|h(x) - h(y)| < \epsilon$.

$$\begin{aligned} |h(x) - h(y)| &= |f(x)g(x) - f(y)g(y)| \\ &= |f(x)g(x) - f(y)g(x) + f(y)g(x) - f(y)g(y)| \\ &\leq |f(x)g(x) - f(y)g(x)| + |f(y)g(x) - f(y)g(y)| \\ &= |g(x)||f(x) - f(y)| + |f(y)||g(x) - g(y)| \\ &\leq N|f(x) - f(y)| + M|g(x) - g(y)| \end{aligned}$$

Let $\epsilon > 0$. Since f is uniformly continuous, $\exists \delta_1 > 0$ such that

$$|x - y| < \delta_1 \implies |f(x) - f(y)| < \frac{\epsilon}{2N}$$

Since g is uniformly continuous, $\exists \delta_2 > 0$ such that

$$|x - y| < \delta_2 \implies |g(x) - g(y)| < \frac{\epsilon}{2M}$$

If we choose $\delta = \min(\delta_1, \delta_2)$ then when

$$|x - y| < \delta \implies |h(x) - h(y)| \leq N|f(x) - f(y)| + M|g(x) - g(y)| < N \cdot \frac{\epsilon}{2N} + M \cdot \frac{\epsilon}{2M} = \epsilon$$

Which is the definition of uniformly continuous, therefore $h(x)$ is uniformly continuous. $\square$

If we drop the assumption that f and g are bounded, then their product is not necessarily uniformly continuous.

Consider $f(x) = g(x) = x$ which is uniformly continuous but not bounded.

Proof:

Choose $\delta = \epsilon$, then when $|x - y| < \delta$, $|f(x) - f(y)| = |x - y| < \delta = \epsilon$.

Their product, $h(x) = x^2$ is not uniformly continuous. Choose $\epsilon = 1$ and let $\delta > 0$.

Consider $x = \frac{1}{\delta}$ and $y = \frac{1}{\delta} + \frac{\delta}{2}$. $|x - y| = |\frac{1}{\delta} - (\frac{1}{\delta} + \frac{\delta}{2})| = \frac{\delta}{2} < \delta$.

$|f(x) - f(y)| = |x^2 - y^2| = |x - y||x + y| = \frac{\delta}{2}|\frac{1}{\delta} + (\frac{1}{\delta} + \frac{\delta}{2})| = (\frac{\delta}{2})(\frac{2}{\delta} + \frac{\delta}{2}) = 1 + \frac{\delta^2}{4} > \epsilon$.

So x^2 is not uniformly continuous since $\exists \epsilon > 0$ such that $\exists \delta > 0$ with $|x - y| < \delta$ but $|f(x) - f(y)| > \epsilon$. □

5 Exercise 5

Find $f'(0)$ if

$$f(x) := \begin{cases} \frac{g(x)}{x} & x \neq 0 \\ 0 & x = 0 \end{cases} \quad (22)$$

and $g(0) = g'(0) = 0$ and $g''(0) = 17$.

5.1 Solution

From the definition of the derivative

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(g(h)/h) - 0}{h} \\ &= \lim_{h \rightarrow 0} \frac{g(h)}{h^2} \text{ which is indeterminate.} \end{aligned}$$

From eq. (9) and eq. (10):

$$\lim_{h \rightarrow 0} \frac{g(h)}{h^2} = \lim_{h \rightarrow 0} \frac{g'(h)}{2h} = \lim_{x \rightarrow 0} \frac{g''(h)}{2} = \frac{17}{2} \quad (23)$$